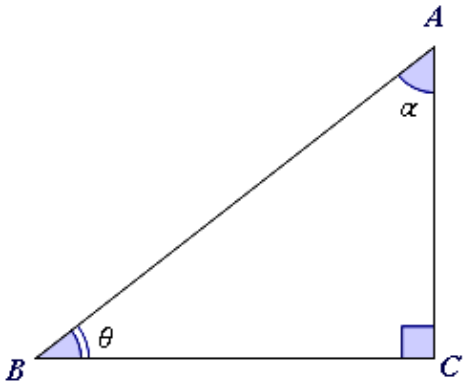
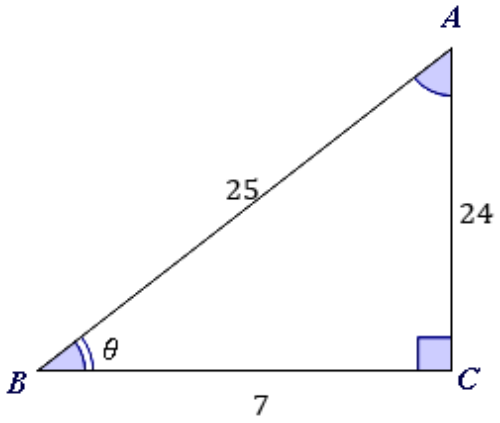
Sine, cosine, and tangent ratios

1 Consider the following right triangle. Identify the angle that is opposite the hypotenuse.

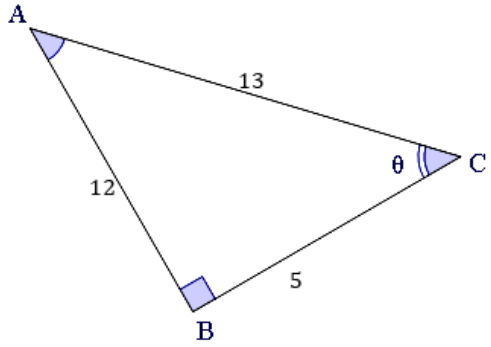


2 Using the reference angle θ, find the value of the ratio $\frac{\text{opposite}}{\text{adjacent}}$ of the following triangles.

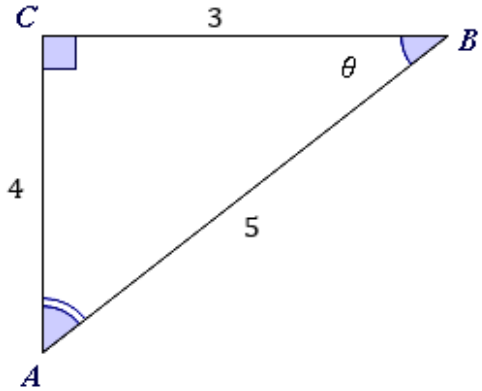
a



b

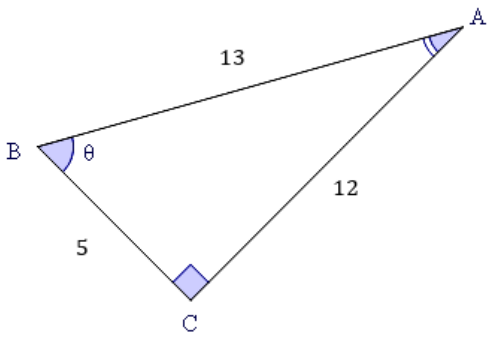


3 Using the reference angle θ, find the value of the ratio $\frac{\text{opposite}}{\text{hypotenuse}}$ for the given triangle.



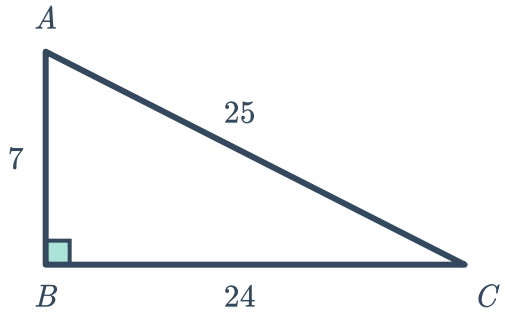
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4 Consider the following right triangle. Using the reference angle θ, find the following ratios:



a $\dfrac{\text{adjacent}}{\text{hypotenuse}}$ **b** $\dfrac{\text{opposite}}{\text{hypotenuse}}$

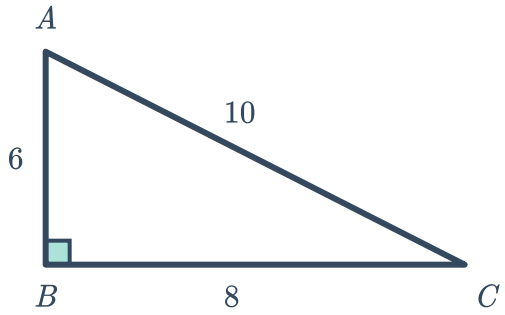
5 Consider the given triangle.



a For $\angle ACB$, find the value of the ratio $\dfrac{\text{opposite}}{\text{hypotenuse}}$.

b For $\angle CAB$, find the value of the ratio $\dfrac{\text{adjacent}}{\text{hypotenuse}}$.

6 Consider the given triangle.

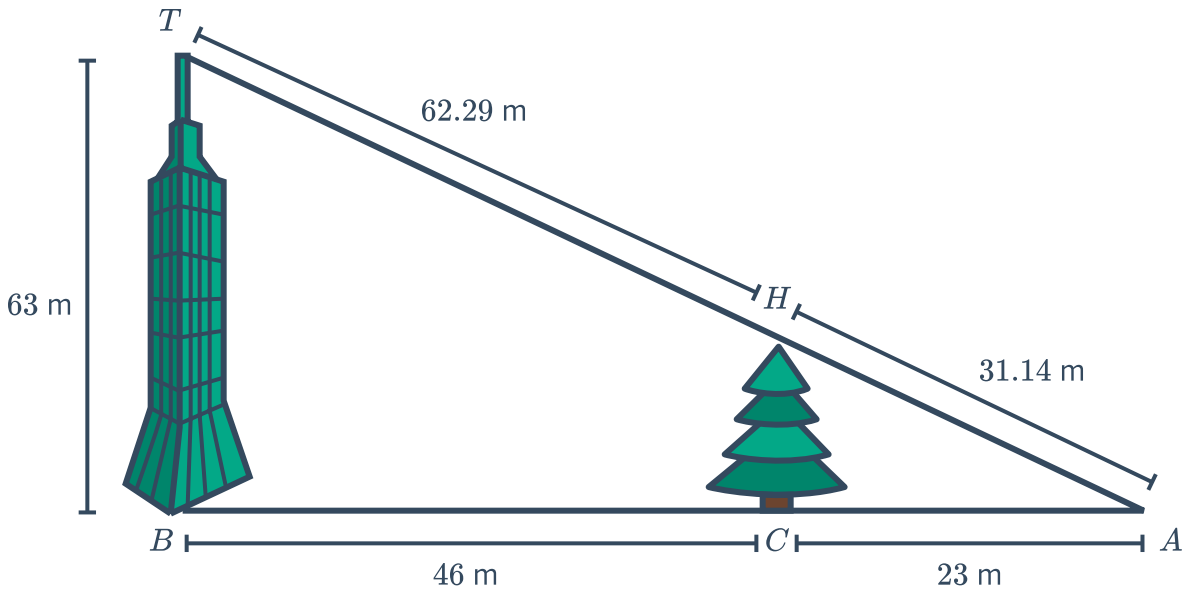


a For $\angle ACB$, find the value of the ratio $\dfrac{\text{adjacent}}{\text{hypotenuse}}$.

b For $\angle CAB$, find the value of the ratio $\dfrac{\text{opposite}}{\text{hypotenuse}}$.

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7 Consider the figure below.



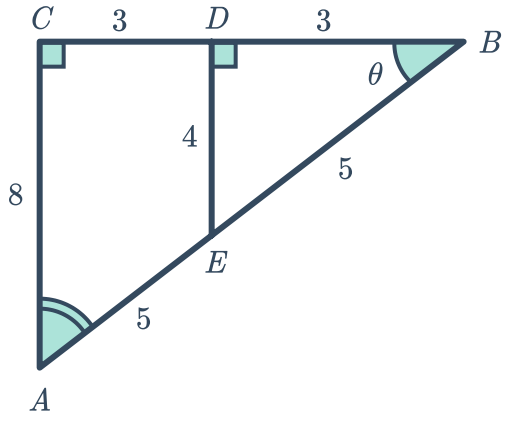
a Using the triangle created by the building, find the ratio of the opposite side to the adjacent side of $\angle TAB$.

b The tree has a height of 21 meters. For the smaller triangle created by the tree, find the ratio of the opposite side to the adjacent side of $\angle HAC$.

c Using the triangle created by the tree, determine the ratio of the adjacent side to the hypotenuse for $\angle HAC$.

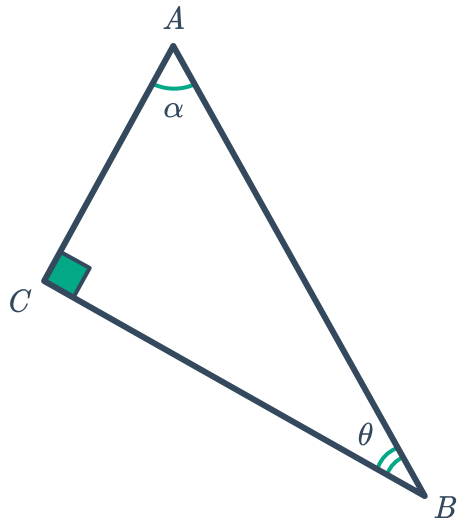
d Using the triangle created by the building, determine the ratio of the opposite side to the hypotenuse for $\angle TAB$.

8 Consider the following diagram.
Using the reference angle θ, state the ratio of sides that is equivalent to $\frac{4}{5}$.



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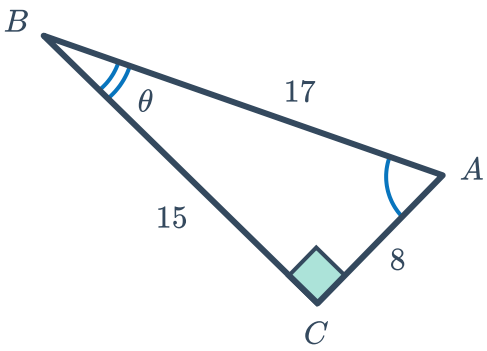
9 Write down the following ratios for the given triangle:



a $\sin\theta$ **b** $\cos\theta$

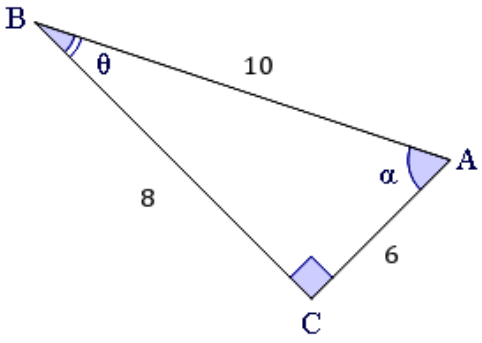
c $\tan\theta$

10 Using the information in the given triangle, find the values of the following ratios:



a $\tan\theta$ **b** $\cos\theta$

11 Using the information in the given triangle, find the values of the following ratios:

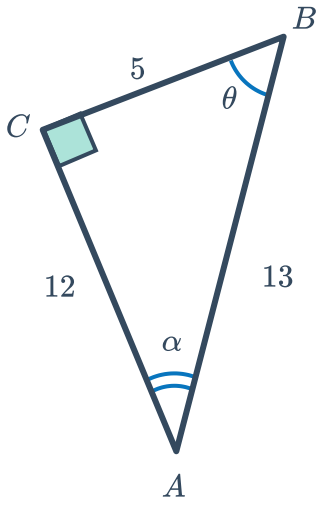


a $\tan\theta$ **b** $\sin\theta$

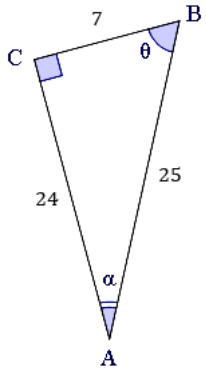
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12 For each of the following triangles, find the value of the given ratio:

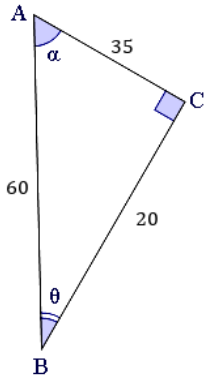
a $\cos\theta$



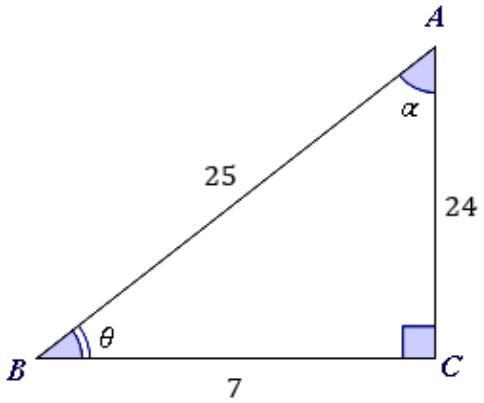
b $\sin\theta$



c $\tan\theta$



d $\tan\theta$

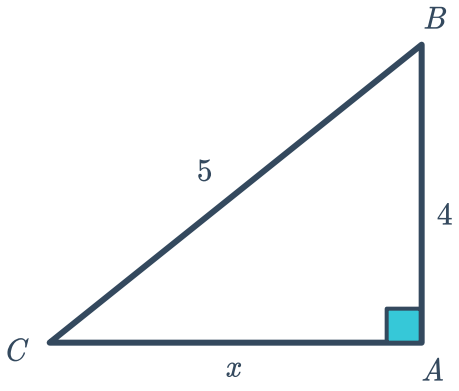


15 In the following triangle, $\sin\theta = \frac{4}{5}$.

a Identify which angle is represented by θ.

b Find the value of $\cos\theta$.

c Find the value of $\tan\theta$.



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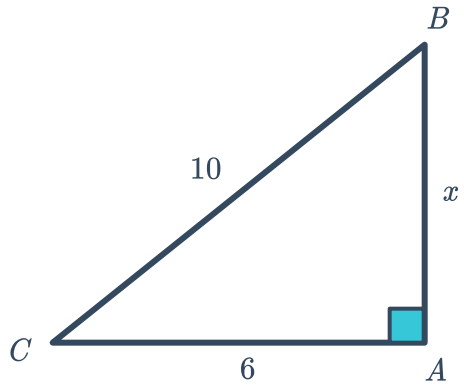
16 For the given triangle, $\cos\theta = \frac{6}{10}$.

a Identify the angle represented by θ.

b Find the value of x.

c Find the value of $\sin\theta$.

d Find the value of $\tan\theta$.

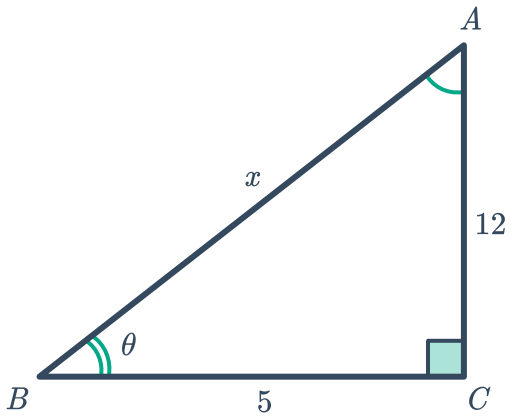


17 Consider the following triangle.

a Find the value of x.

b Find the value of $\sin\theta$.

c Find the value of $\cos\theta$.



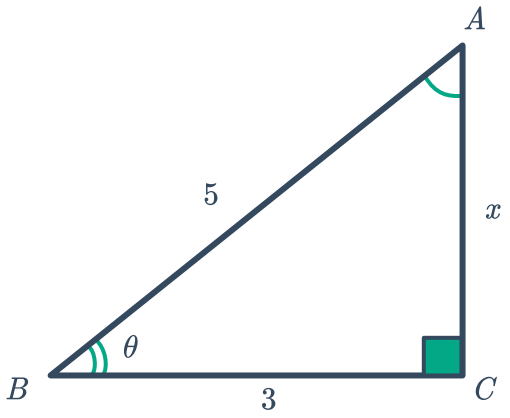
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18 For each of the following triangles:

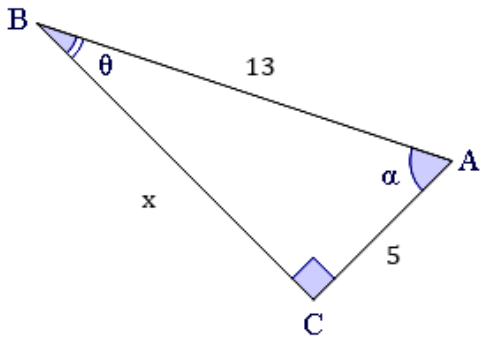
i Find the value of x.

ii Hence, find the value of $\tan\theta$.

a



b



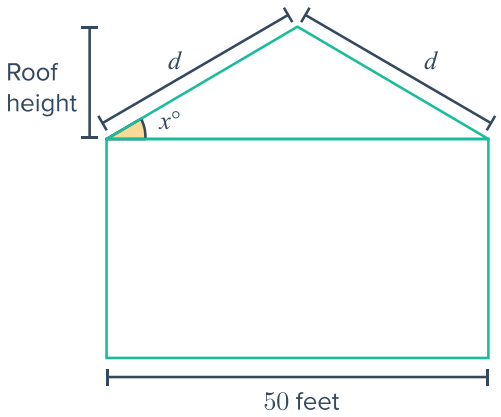
19 Determine whether the following pairs of right triangles are similar.

a A pair of corresponding angles are found to have a sine ratio of $\frac{240}{250}$ and $\frac{24}{25}$.

b A pair of corresponding angles are found to have a tangent ratio of $\frac{140}{160}$ and $\frac{70}{80}$.

c A pair of corresponding angles are found to have a tangent ratio of $\frac{320}{420}$ and $\frac{316}{416}$.

20 Beth is building a house, as shown.
She wants the roof height to be between 3.5 ft and 5 ft. She must decide the angle measure to use for the slant of the roof when the slant height is d ft.
State the inequality that Beth can use in order to ensure that her roof height will be within the desired range.



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Additional questions

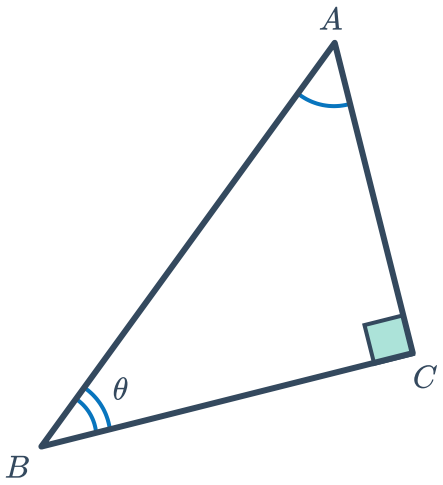
21 For the following triangles, with reference angle θ, identify:

i The adjacent side

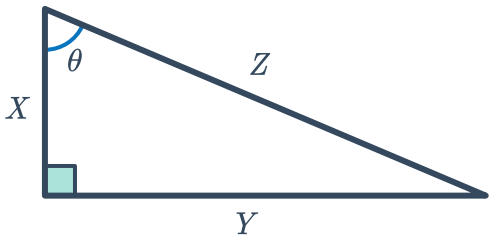
ii The opposite side

iii The hypotenuse

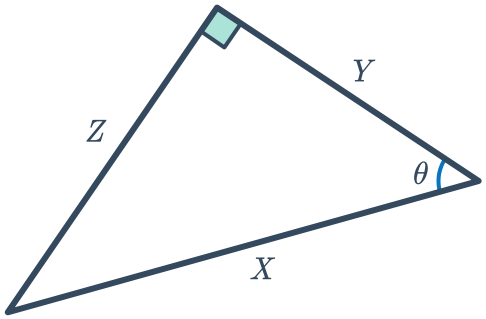
a



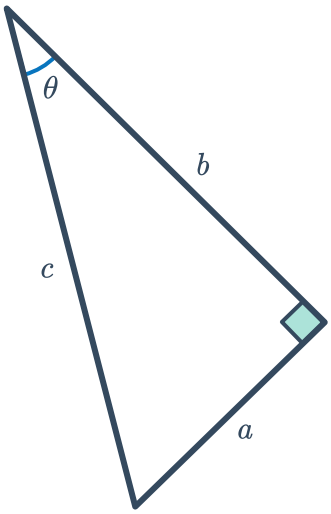
b



c



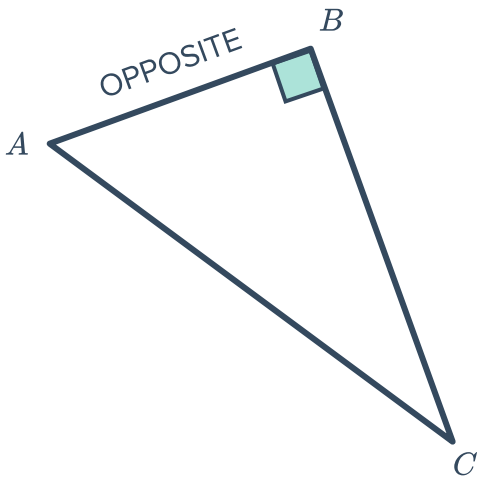
d



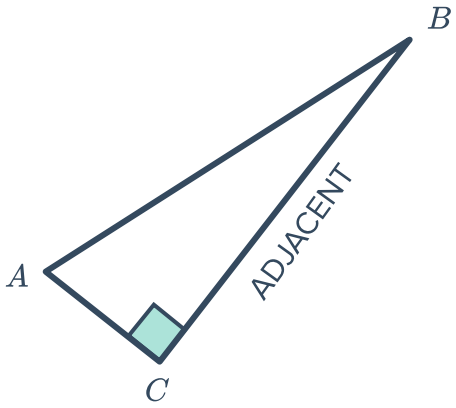
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22 Identify the reference angle in each of the following triangles given the labelled side:

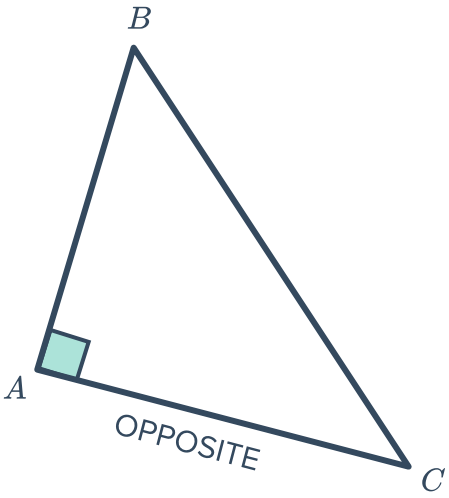
a



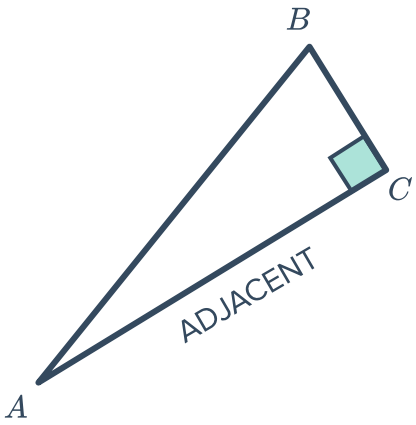
b



c



d



23 Consider the following triangle:

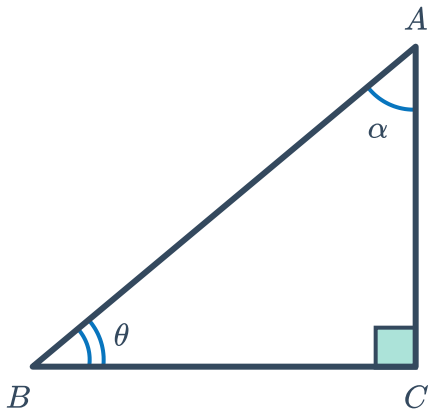
a State the opposite side to angle θ.

b State the adjacent side to angle θ.

c State the opposite side to angle α.

d State the adjacent side to angle α.

e Identify the angle that is opposite the hypotenuse.



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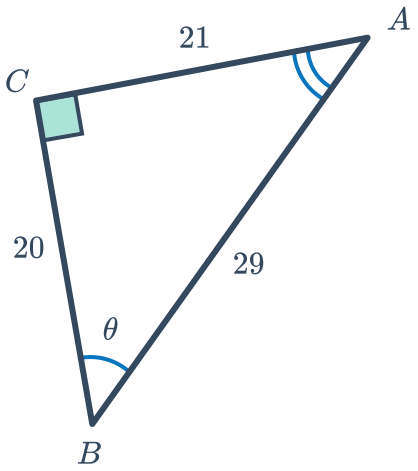
24 For each of the following triangles:

i Find the value of $\sin\theta$.

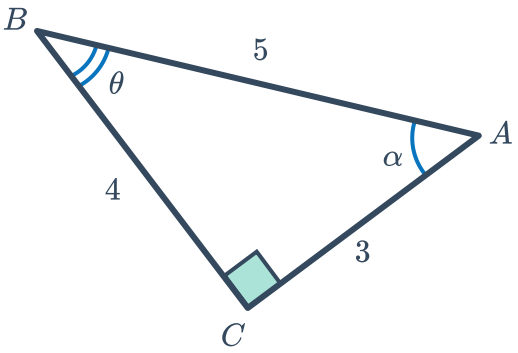
ii Find the value of $\cos\theta$.

iii Find the value of $\tan\theta$.

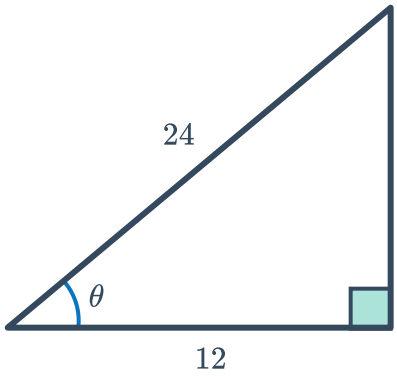
a



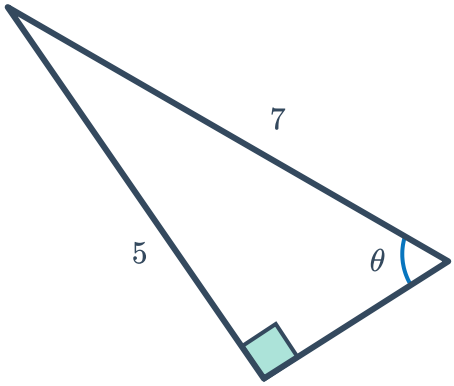
b



c

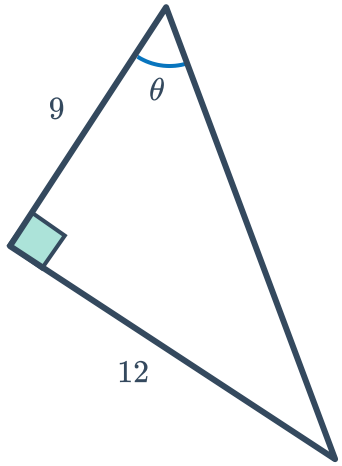


d

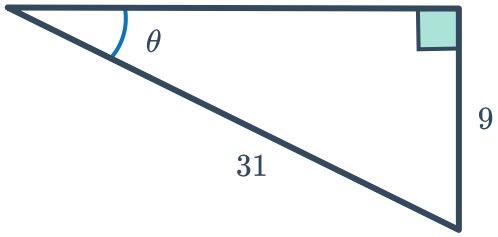


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e



f

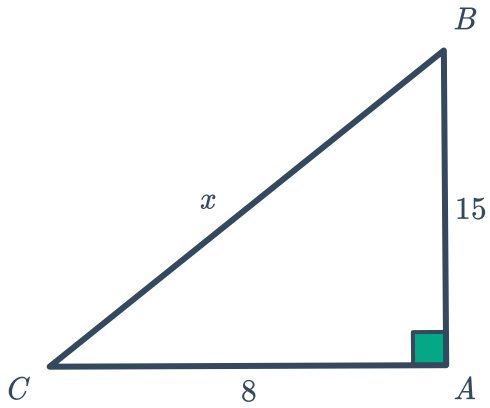


25 In the following triangle $\tan\theta = \frac{15}{8}$.

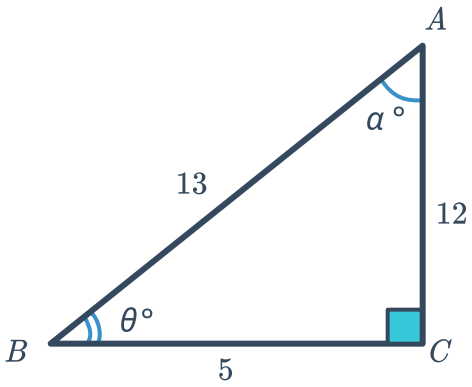
a Which angle is represented by θ?

b Find $\cos\theta$.

c Find $\sin\theta$.

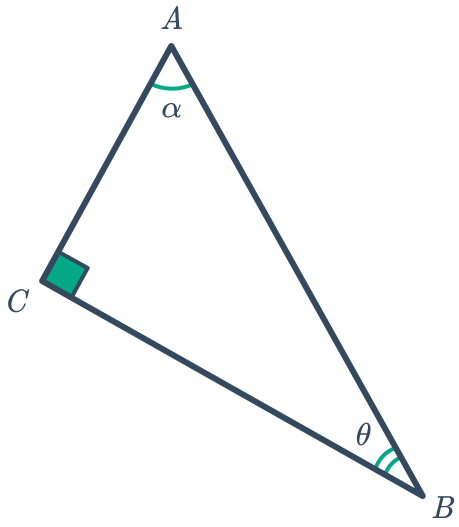


26 With reference to the diagram, show that $\frac{\sin\theta}{\cos\theta} = \tan\theta$.



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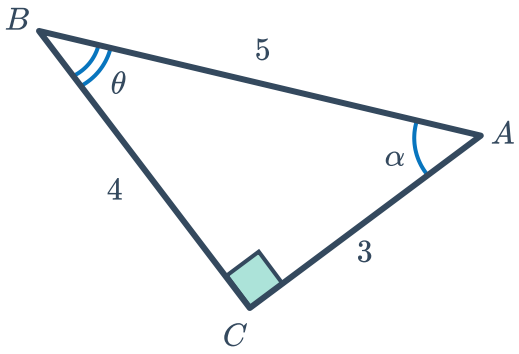
27 Write down the following ratios for the given triangle:



a $\sin\alpha$ **b** $\cos\alpha$

c $\tan\alpha$

28 With reference to the diagram, does $\sin\theta = \cos\alpha$? Explain why or why not.



29 A right triangle is enlarged by a factor of 4. Explain why the trigonmetric ratios are the same for both triangles.

30 Explain why $\tan 30^\circ = \dfrac{\sqrt{3}}{3}$. Include a diagram with your explanation.

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